Earth and Planetary Science Letters, 85 (1987) 488–496
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

Magnetic polarity stratigraphy of Upper Siwalik Sub-Group, Soan Valley, Pakistan: implications for early human occupance of Asia

H.M. Rendell [1], E.A. Hailwood [2] and R.W. Dennell [3]

[1] *Geography Laboratory, University of Sussex, Falmer, Brighton BN1 9QN (U.K.)*
[2] *Department of Oceanography, University of Southampton, Southampton SO9 5NH (U.K.)*
[3] *Department of Archaeology and Prehistory, University of Sheffield, Sheffield S10 2TN (U.K.)*

Received March 24, 1987; revised version accepted July 18, 1987

The results of a detailed paleomagnetic study of a 68 m section of Upper Siwalik sediments in the Soan syncline, northern Pakistan, are presented. A palaeolithic artefact and other pieces of struck quartzite were found in situ in a gritstone/conglomerate horizon near the base of the section. Incremental thermal demagnetization was used to remove later magnetic overprints in these sediments, since alternating field demagnetization was shown to be inappropriate. With the exception of the lowest stratigraphic level, the Upper Siwalik sediments examined in the Riwat section show reverse polarity magnetization. The declination values are consistent with a 16° (±4°) counterclockwise rotation of the Soan syncline tectonic block since deposition of the sediments. On the basis of the palaeomagnetic analyses and the tectonic and stratigraphic context of the section, our current best estimate of the age of the artefact-bearing horizon is 2.0 ± 0.2 Ma.

1. Introduction

The Siwalik Series of fluvial molasse deposits exposed in the Potwar area of northern Pakistan present an unparalleled opportunity for studies of foredeep sedimentation. These molasse deposits preserve a 17 million year record of the Himalayan orogeny. Foredeep sedimentation was sequentially disrupted during the Late Pliocene and Pleistocene [1]. Recent studies of the magnetic polarity stratigraphy of the Siwaliks, coupled with fission track dating of volcanic tuff horizons has enabled the construction of a broad chronological framework for Siwalik sedimentation and tectonic disruption [2–4].

The long sequences of Siwalik sediments present an excellent opportunity for study of the palaeomagnetic record of terrestrial sediments. The deposition rates are generally high, of the order of 0.2–0.5 m/ka [5], and these rates allow detailed studies of magnetization to be undertaken, using close sampling intervals, equivalent to relatively short periods of geological time. Sequences are not strictly continuous, erosional breaks being common, particularly at the base of lateral accretion deposits. Sampling is also limited largely to the fine-grained vertical accretion deposits since the sands generally are too weakly magnetized to yield reliable palaeomagnetic data. In some areas, where sections are dominated by fine-grained sediments, as is the case in the present study, sampling at intervals of about 0.5 m throughout the section becomes feasible.

We present the preliminary results of a detailed study of a 68 m section of Upper Siwalik sediments outcropping along the southeastern limb of the Soan syncline. The section lies to the west of the Grand Trunk Road and 4.8 km north of Riwat (33°31′N, 73°9′E). The importance of the section lies not only in the inferences that can be drawn from the palaeomagnetic study of the sediments but also in the fact that a stone artefact and other pieces of struck stone were found in situ near the base of the section. Finds of stone artefacts in situ in the Siwaliks have been reported previously from the eastern Potwar [6], but, given the stratigraphic and tectonic setting, the Soan context is likely to be of greater antiquity. The section has significant archaeological importance, and the measurements of magnetic polarity stratigraphy have the potential to provide a chronological framework for these artefacts.

0012-821X/87/$03.50 © 1987 Elsevier Science Publishers B.V.

2. Siwalik stratigraphy

The Siwalik Series is characterised by striking lateral changes in facies which have presented a severe challenge to all geologists who have attempted to map these deposits. Stratigraphic frameworks were originally established on the basis of studies of vertebrate faunas, principally by Pilgrim [7]. Later workers established lithostratigraphic units and these units form the current basis for Siwalik stratigraphy [8]. The uppermost Siwalik sediments were deposited during a time of tectonic disruption of foredeep sedimentation. Not surprisingly, the Upper Siwalik Sub-Group exhibits rapid lateral facies changes and, in the absence of faunal data, has proved exceptionally difficult to map. The establishment of magnetic polarity stratigraphies for this Sub-Group has been extremely valuable and has enabled inferences to be made about sedimentation rates and depocentre migration. Furthermore, these stratigraphies have provided a time-frame for the tectonic disruption of foredeep sedimentation during Upper Siwalik times [1,5].

3. The Soan syncline

Some 4880 m of Siwalik strata are exposed in the Soan syncline, 3–10 km to the southeast of Rawalpindi, of which 1700 m have been mapped as Upper Siwalik [9]. The synclinal trough is strongly asymmetrical with its axis running northeast-southwest (Fig. 1). It comprises a steeply dipping to vertical northwestern limb and a gently dipping southeastern limb. The syncline is bounded to the north by a complex folded series

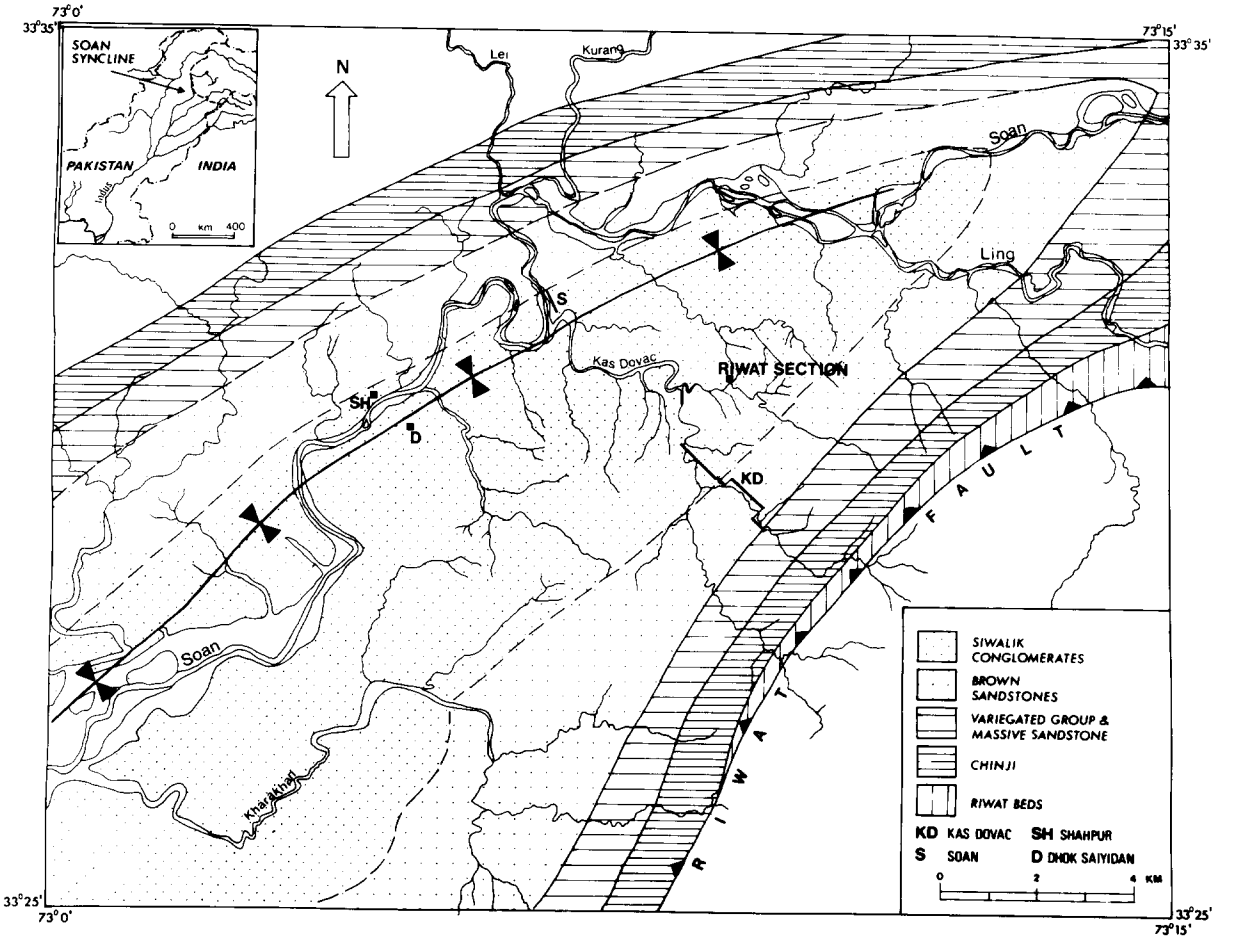


Fig. 1. Location of key sections within the Soan syncline. Stratigraphy after Gill [9].

of beds belonging to the Murree Group and to the southeast by the Riwat Fault. The Upper Siwalik deposits exposed in the syncline comprise alternating beds of sandstone, pinkish-red siltstone and claystone with some massively bedded quartzite-rich conglomerates [8,9].

Recent work on the magnetic polarity stratigraphy of two sections of Siwalik sediments and one section of post-Siwalik sediments within the Soan syncline area has provided a chronological fix on the deformation of the syncline. Siwalik sections on the northwestern and southeastern limbs of the fold at Soan and Dhok Saiyidan, respectively, are reported by Moragne [10] and Raynolds and Johnson [5]. Moragne measured a 650 m section along the Soan river just to the north of the confluence with the Kas Dovac. The dip of the beds at this section is almost vertical. A volcanic ash outcrops at about 110 m below the top of the section and coincides with a change in polarity in the section from normal polarity below to reverse polarity above which has been correlated with the Gauss-Matuyama Chron boundary. The same ash horizon was traced to the southern limb of the syncline at Dhok Saiyidan where the beds dip at 9° towards the synclinal axis. At Shahpur, across the Soan river from Dhok Saiyidan, Moragne reported a horizontal sequence of sediments overlying the folded Siwalik beds with a strong angular unconformity. The unconformable sequence contains a bentonitised volcanic ash which has been dated by the fission track method to 1.60 ± 0.18 Ma [4]. This ash coincides with a normal polarity event within reversely magnetised sediments which has been identified as the Olduvai Subchron [5, fig. 17]. Given these data, the youngest age for Siwalik sediments within the Soan syncline appears to be ca. 1.9 Ma. The northwestern part of the area was strongly compressed, with beds being tilted close to the vertical and was then subject to a massive amount of erosion before deposition of the horizontal ash-bearing sequence at about 1.6 Ma. Raynolds [11] argues that the tectonic evolution of the Soan area comprised two phases. First, during the interval 3.5–2.5 Ma, the Riwat Fault was active, uplifting the southern part of the area and accounting for a gap in the sedimentary sequence on the southern limb of the fold during the lower Gauss Chron. This movement had ceased by ca. 2.4 Ma. Second, during the interval 2.1–1.9 Ma, the main folding event of the syncline occurred. Although deformation appears to have ceased by 1.6 Ma, some of the palaeomagnetic data presented by Burbank and Raynolds [1] suggest that the syncline has undergone a 10–20° counterclockwise rotation between about 1.8 Ma and the present.

In this paper we present new palaeomagnetic data from a 68 m sequence of Siwalik sediments (the Riwat section) exposed on the right bank of a tributary to the Kas Dovac (see Fig. 1). The section is on the southern limb of the syncline and the strata at the base of the section dip at 12–13° towards the synclinal axis, whereas the dip of the uppermost beds is 4–5°. A quartzite artefact and 10 other pieces of struck quartzite were found in situ in the gritstone/conglomerate horizon close to the base of the section between sampling levels 2 and 3A. Two orientated samples were taken at each of 45 stratigraphic levels in the section. At least two 8 cm^3 cubes were cut from each orientated sample, yielding a total of at least four palaeomagnetic specimens per sampling level. Sampling was limited to the fine-grained vertical accretion deposits within the profile.

4. Palaeomagnetic results

4.1. Stability of magnetization and magnetic polarity

The first step in the palaeomagnetic analyses involved measuring the natural remanent magnetization (NRM) of all samples. The variation of NRM intensity with stratigraphic height is shown in Fig. 2. In general the values lie in the range 0–10 mA/m but significantly higher values occur at certain points in the section, particularly levels 1, 23, 34 and 44. These fluctuations in intensity of magnetization most probably represent variations in the type and concentration of magnetic minerals. The intensity peak at site 23 is particularly conspicuous, the mean value (82.7 mA/m) being nearly two orders of magnitude greater than the mean for the rest of the section (3.52 mA/m). This requires a local concentration of ferromagnetic material at this level, and one possible explanation is the presence of a thin bentonite layer. Mineralogical studies of the specimens from this level are currently in progress to explore this possibility.

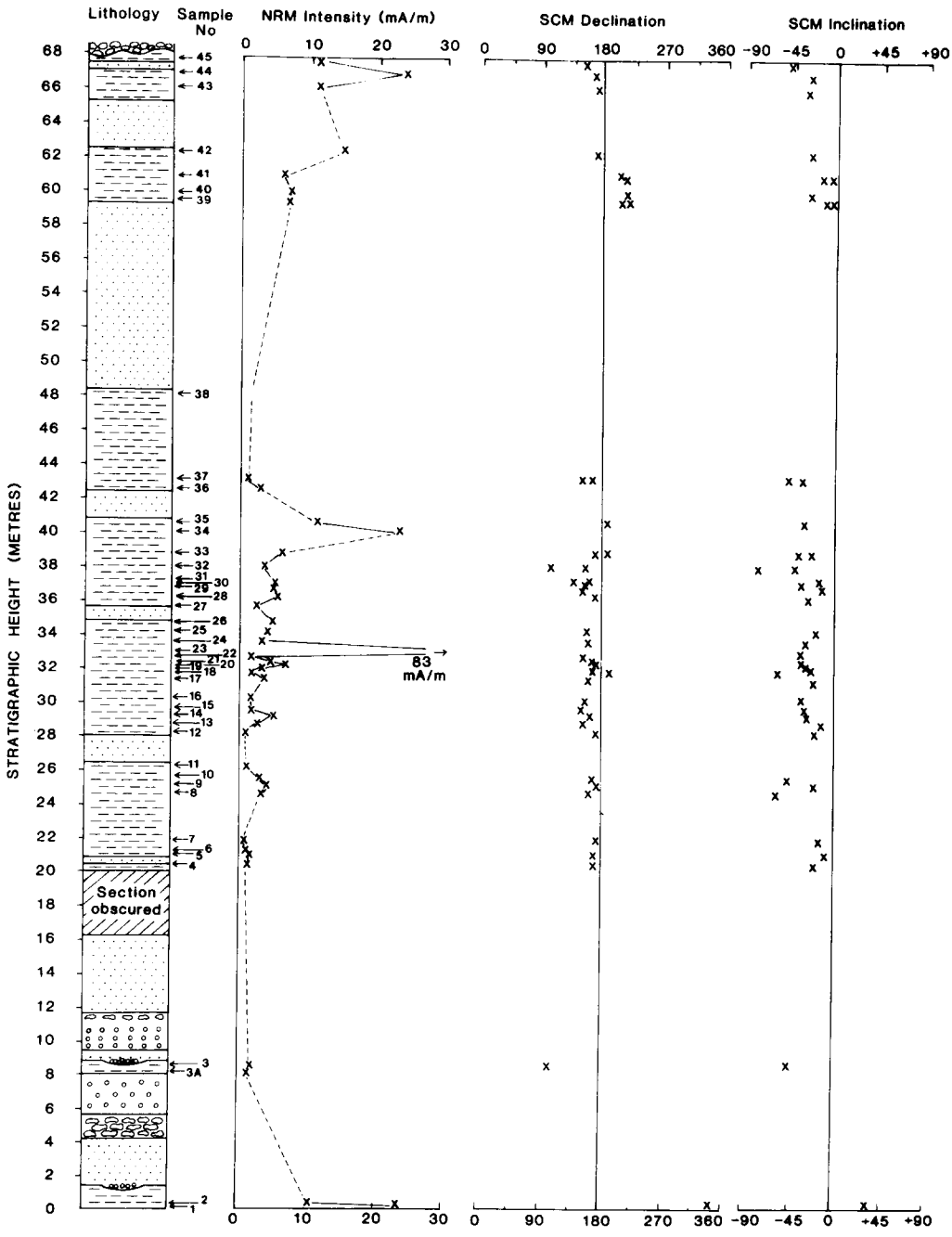


Fig. 2. Lithological summary and location of sampling levels in the Riwat section. Also shown is the stratigraphic variation of intensity of natural remanent magnetization and the declination and inclination of the stable characteristic magnetization (SCM) identified by thermal demagnetization in category A samples: in the lithological column dots represent sand units and dashes silt-clay beds.

In order to investigate the stability of the remanent magnetization within these sediments and to resolve the separate components in samples carrying a multicomponent remanence, specimens from each stratigraphic level were subjected to incremental demagnetization analysis. To establish the

optimum demagnetization procedure, initially pairs of specimens were selected from each of six stratigraphic levels. One specimen from each pair was subjected to incremental alternating field (AF) demagnetization and the other to thermal demagnetization.

The AF demagnetization was carried out at steps of 2.5 or 5.0 mT and the thermal demagnetization at steps of 50 °C until the magnetization became too weak for reliable determinations to be made with the Molspin magnetometer used in this study. This point was normally reached at applied fields of 40 or 50 mT or temperatures of 550–600 °C. The responses of representative samples to this treatment are illustrated in Fig. 3.

Specimens from level R1 show similar responses to AF and thermal treatment. In both cases the vector maintains a NNW declination and intermediate inclination throughout demagnetization and the points on the vector end point (VEP) diagrams define one or more linear segments. However, whereas the final linear segment for the thermal demagnetization plot is directed through the origin (Fig. 3a), that for the AF demagnetization plot is not (Fig. 3b). Thus, in this sample, thermal demagnetization appears to be more effective in defining a demagnetization stable end-point, although both types of treatment indicate that the stable characteristic magnetization (SCM) at this level is aligned in the same

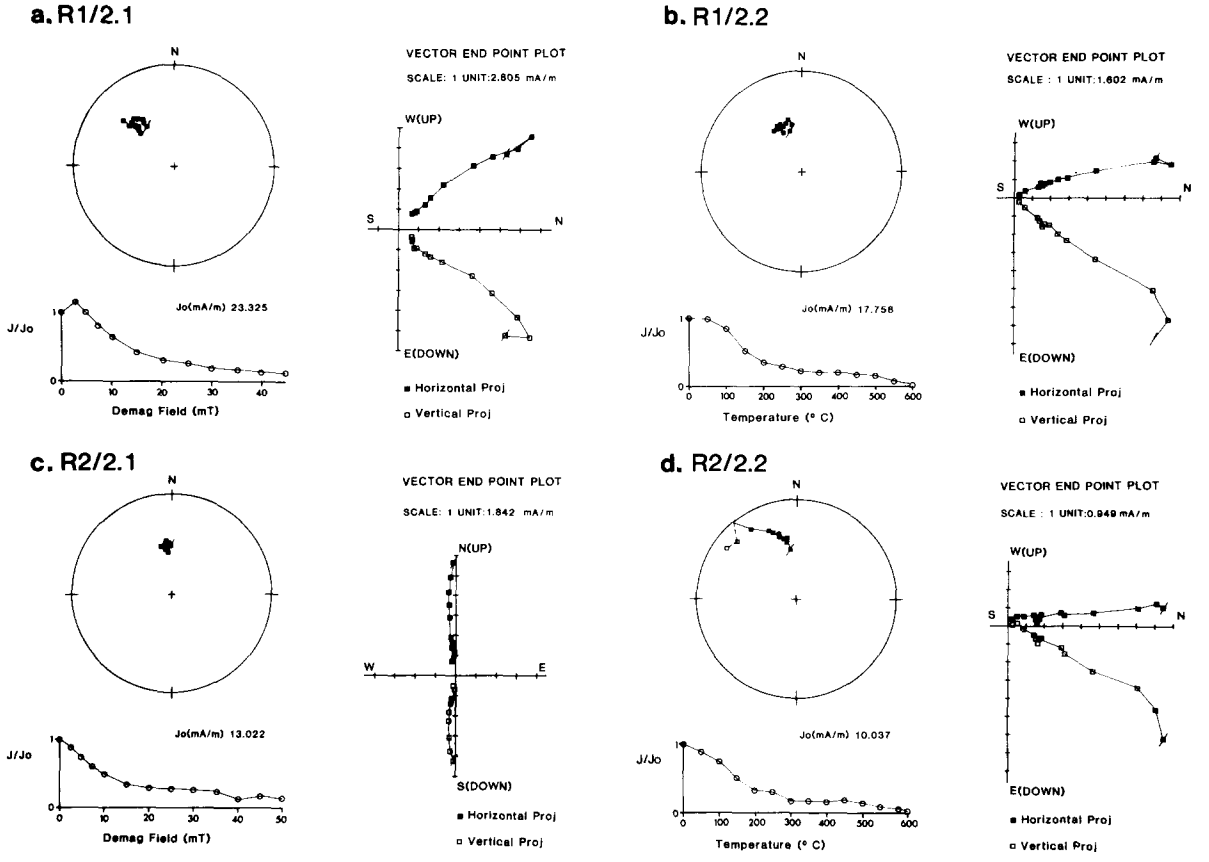


Fig. 3. Comparison between alternating field (AF) and thermal demagnetization behaviour of pairs of samples from two stratigraphic levels. (a) and (b) illustrate the identification of a normal polarity magnetization by both types of treatment for Site 1. (c) and (d) indicate how thermal demagnetization reveals the presence of a reverse polarity component which is not evident from AF treatment. In all four figures the left-hand diagrams show an equal-area stereographic projection of the magnetization direction and a graph of the magnetization intensity (J) normalized by the NRM value (J_0), after each demagnetization step. The right-hand diagram shows projections of the vector end-point onto a vertical and a horizontal plane using the convention of Zijderveld [15].

sense as the present geomagnetic field (northerly declination, positive inclination) and therefore represents a normal polarity.

In contrast, the pair of specimens from level R2 show different responses to the two types of demagnetization. The directions of magnetization remain close to the present geomagnetic field during AF demagnetization (Fig. 3c) whereas during thermal treatment (Fig. 3d) they undergo a progressive change towards a shallow negative vector. This directional trend can be extrapolated to a reversed polarity (southerly declination, negative inclination) end point. This behaviour indicates the probable presence of a reverse polarity component of magnetization, carried by magnetic grains with high blocking temperature. Neither type of demagnetization completely isolates this component. However, its presence can be deduced from the thermal demagnetization data but not from the AF data.

The other four pairs of trial specimens showed broadly similar behaviour to those from level R2 and it was concluded that thermal demagnetization was the most appropriate technique for defining the direction of the SCM in these sediments. In consequence, specimens from all other levels were subjected to incremental thermal treatment at 50 °C intervals up to maximum temperatures in the range 600–650 °C.

The results of this treatment allow the palaeomagnetic data from each of the samples to be placed in four broad reliability categories. These categories are illustrated in Fig. 4 and described below.

Category A. After removal of a weak viscous component of magnetization, a well-established demagnetization stable end-point is defined by a linear trend through the origin of the VEP plot (e.g. specimen R24/2.1, Fig. 4a). The polarity is established beyond question. The specific direction of the SCM is regarded as being reliable to better than 5°.

Category B. A linear trend through the origin of the VEP is evident after thermal demagnetization above about 200 °C but this is less well-defined than for category A vectors (e.g. specimen R22/1.1, Fig. 4b). The polarity of the SCM is established beyond question but the specific magnetization direction is only reliable to about 10°.

Category C. A clear directional trend occurs towards a reverse polarity demagnetization stable end-point, but this point is either not completely

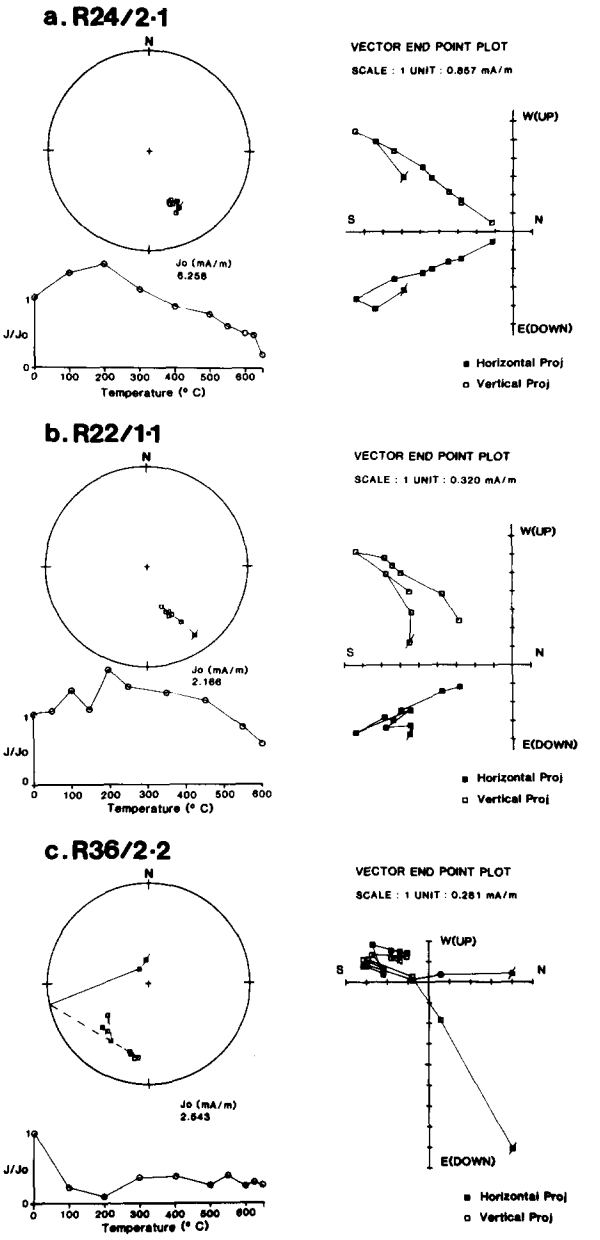


Fig. 4. Illustrations of typical response to thermal demagnetization of (a) category A, (b) category B, and (c) category C samples. Symbols and conventions as in Fig. 3.

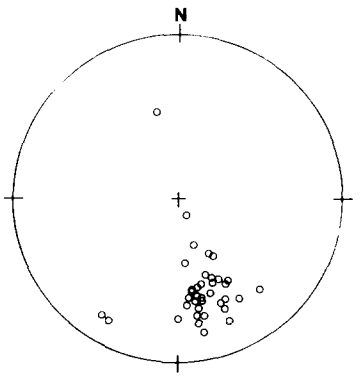


Fig. 5. Equal-area stereographic projection of stable magnetization vectors identified by thermal demagnetization in category A samples.

reached or is partially obscured by instrumental noise (e.g. specimen R36/2.2, Fig. 4c). The polarity of the SCM is established beyond question but the specific magnetization direction is only poorly defined, with an uncertainty $> 10°$.

Category D. Either the intensity of NRM is too weak or the demagnetization data are too noisy to permit a reliable definition of the polarity.

Where the behaviour of specimens appeared to fall between two classes they were downgraded to the lower class. Out of 82 specimens analysed, 56 were assigned to category A, 7 to category B, 17 to category C and 2 to category D. The tilt-corrected SCM declination and inclination values defined by thermal demagnetization for each of the category A specimens are plotted as a function of stratigraphic height in Fig. 2 and the vectors are shown on an equal-area stereographic projection in Fig. 5. In the latter projection the data for each stratigraphic level have been combined into a mean vector for that level before plotting. With the exception of the lowermost site, all other category A sites are characterised by a reverse polarity stable magnetization, as are all category B and C sites. It is concluded that virtually the whole of the Riwat section was deposited during a reverse polarity magnetic chron.

4.2. Specific direction of stable magnetization

A striking feature of the stable magnetization vectors (Fig. 5) is that they deviate systematically to the east of the expected southerly mean for a reverse polarity. The overall mean vector has a declination of 166° when all data points are included and 164° when the more dispersed data points (those which deviate more than 2 standard deviations from the mean) are eliminated (Table 1). In both cases the declination is different at the 95% confidence level (6° and 4°, respectively) from the expected declination of 180°. This result may be explained in two ways. Either the interval of time represented by this sequence of sediments is too short for geomagnetic field secular variations to have been properly averaged out, or else this region must have undergone a major counterclockwise rotation through some 15° since the sediments were deposited. As regards the first possible explanation, a typical mean sedimentation rate for Siwalik sediments is believed to be in the range 0.2–0.5 m/ka [3]. Thus the total time interval represented by this 68 m section is likely to be of the order of at least 120 ka. It is widely believed that the minimum time required to satisfactorily average out the geomagnetic secular variation so that the mean palaeomagnetic vector gives a reliable estimate of the geographic north (or south) is ca. 10^4 years [12]. For this reason we believe that the easterly departure of the mean vector for the Riwat section from due south is best explained by a counterclockwise rotation of the region through a total angle of about 16° (±4°) since deposition of the sediments.

The mean inclination of the stable characteristic magnetization for the Riwat section, after correction for structural dip, is −29° whereas the expected value for the present-day latitude of the sampling site is −53°. The discrepancy cannot be accounted for by the northward movement of the Indian plate, which is unlikely to have exceeded a few degrees at most since deposition of the sedi-

TABLE 1

Mean palaeomagnetic directions (category A vectors)

	D (°)	I (°)	α_{95} (°)	CSD	k	N
Group 1	166	−30	6	23	12.2	56
Group 2	164	−29	4	15	30.2	47

Group 1. All sites

Group 2. After elimination of all vectors which depart by more than 2×CSD from the mean of the other vectors.

α_{95} is the radius of the 95% confidence circle about the mean, CSD is the circular standard deviation, k is the precision parameter [16] and N is the number of vectors used in the calculation.

ments. The most likely explanation is the presence of a depositional inclination error, which is prone to occur in rapidly-deposited clastic sediments [13,14].

5. The stratigraphic context of the Riwat section

Stratigraphically the Riwat section comprises some of the youngest Siwalik sediments exposed in the Soan syncline. The section lies above Raynolds' Kas Dovac section [11], and palaeomagnetic investigation of the stratigraphic interval between the two sections is now in progress by the present authors. The uppermost portion of Raynolds' section is interpreted as belonging to the Gauss Chron. If the normal polarity zone identified in the single site at the base of the Riwat section is confirmed, then the youngest possible age for this normal polarity interval would be ca. 2.1 Ma (i.e. the Reunion event), since the younger Olduvai Subchron has been identified in the overlying horizontally-bedded ash-bearing post-Siwalik sediments at Shahpur [5]. Given that the youngest age for Siwalik sediments in the Soan syncline is 1.9 Ma, our current best estimate for the age of the artefact-bearing horizon at the base of the Riwat section is 2.0 ± 0.2 Ma.

6. Tectonic context of the Riwat section

Further corroborative evidence for the age of the section is provided by the palaeomagnetic indication of a counterclockwise rotation of the Soan syncline tectonic block since deposition of the sediments. The existence of such a rotation was postulated by Burbank and Raynolds [1] who estimate that a rotation of some 20° (with an uncertainty of about $\pm 15°$) has occurred between 1.8 Ma and the present. Our results support this postulated rotation and help to constrain the total angle. All of our measurements are based on incrementally demagnetized data and indicate a total rotation of 16° with 95% confidence limits of 4° (Table 1).

7. Conclusions

The Upper Siwalik sediments examined in the Riwat section exhibit reverse polarity magnetization, with the exception of only one sample, at the base of the section, which shows a probable normal polarity. The results emphasise the need for incremental thermal demagnetization to remove later magnetic overprints in these sediments. Alternating field demagnetization has been shown to be inappropriate. Measurements of NRM intensities down the section have identified the possible existence of a volcanic ash at level 23. The palaeomagnetic data show inclination values which are too shallow to be consistent with the present latitude of the section, and may be explained by the presence of a depositional inclination error. The majority of declination values are oriented to the SSE, and are consistent with a 16° counterclockwise rotation of the Soan Syncline tectonic block since deposition of the sediments.

Given the stratigraphic and tectonic setting of the section, the sediments may be assigned to the early (Pre-Olduvai) portion of the Matuyama reverse polarity chron. We estimate that the youngest possible age for the artefact-bearing horizon is 2.0 Ma. This finding has profound implications for studies of early man in Asia, since it puts the date of human occupance back 1 Ma earlier than all previous estimates.

Acknowledgements

Fieldwork was undertaken as part of the Palaeolithic Project of the British Archaeological Mission to Pakistan, with financial support from the British Academy and the Royal Society. We thank the Directors of the British Mission, Drs. F.R. and B. Allchin, the Director-General of the Government of Pakistan Department of Archaeology and the Director-General of the Geological Survey of Pakistan, Dr. Ibrahim Shah (Deputy Director-General G.S.P.) and their colleagues for their invaluable support. The field assistance of Saleem Ul-Haq and Eddie Moth is gratefully acknowledged. The laboratory work was funded by the British Archaeological Mission to Pakistan and the laboratory measurements were made with great care and efficiency by Dave Crawford.

References

1 D.W. Burbank and R.G.H. Raynolds, Sequential Late Cenozoic structural disruption of the northern Himalayan foredeep, Nature 311, 114–118, 1984.

2 N.D. Opdyke, G.D. Johnson, N.M. Johnson, R.A.K.

Tahirkheli and M.A. Mirza, Magnetic polarity stratigraphy and vertebrate palaeontology of the Upper Siwalik Subgroup of northern Pakistan, Palaeogeogr., Palaeoclimatol., Palaeogeocol. 27, 1–34, 1979.

3 N.M. Johnson, N.D. Opdyke, G.D. Johnson, E.H. Lindsay and R.A.K. Tahirkheli, Magnetic polarity stratigraphy and ages of Siwalik Group rocks of the Potwar Plateau, Pakistan, Palaeogeogr., Palaeoclimatol., Palaeoecol. 37, 17–42, 1982.

4 G.D. Johnson, P.K. Zeitler, C.W. Naeser, N.M. Johnson, D.M. Summers, C.D. Frost, N.D. Opdyke and R.A.K. Tahirkheli, Fission track ages of Late Neogene and Quaternary volcanic sediments, Siwalik Group, northern Pakistan, Palaeogeogr., Palaeoclimatol., Palaeoecol. 37, 63–93, 1982.

5 R.G.H. Raynolds and G.D. Johnson, Rates of Neogene depositional and deformational processes, north-west Himalayan foredeep margin, Pakistan, in: The Chronology of the Geological Record, N.J. Snelling, ed., pp. 297–311, Geological Society, London, 1985.

6 H.M. Rendell and R.W. Dennell, Lower palaeolithic artefacts dated in northern Pakistan, Current Anthropol. 26(3), 393, 1985.

7 G.E. Pilgrim, Preliminary note on a revised classification of the Tertiary freshwater deposits of the North-West Punjab, Rec., Geol. Surv. India 40, 185–205, 1910.

8 A.N. Fatmi, ed., Lithostratigraphic units of the Kohat-Potwar Province, Indus Basin, Pakistan, Mem., Geol. Surv. Pakistan, 10, 1973.

9 W.D. Gill, The stratigraphy of the Siwalik Series in the northern Potwar, Punjab, Pakistan, Q.J. Geol. Soc. London 107, 395–421, 1952.

10 J.H. Moragne, Magnetic polarity stratigraphy and timing of deformation for an Upper Siwalik sedimentary sequence, Soan syncline, Pakistan, 31 pp., B.Sc. Thesis, Dartmouth College, Hanover, N.H., 1979.

11 R.G.H. Raynolds, The Plio-Pleistocene structural and stratigraphic evolution of the eastern Potwar Plateau, Pakistan, 265 pp., Ph.D. Thesis, Dartmouth College, Hanover, N.H., 1980.

12 D.H. Tarling, Palaeomagnetism: Principles and Applications in Geology, Geophysics and Archaeology, 379 pp., Chapman and Hall, London, 1983.

13 R.F. King, The remanent magnetism of some artificially deposited sediments, Mem., R. Astron. Soc. Geophys., Suppl. 7, 115–134, 1955.

14 R.A. Blow and N. Hamilton, Palaeomagnetic evidence from DSDP cores of northward drift of India, Nature 257, 570–572, 1975.

15 H.A. Zijderveld, A.C. demagnetization of rocks: analysis of results, in: Methods in Palaeomagnetism, D.W. Collinson, K.M. Creer and S.K. Runcorn, eds., pp. 254–286, Elsevier, Amsterdam, 1967.

16 R.A. Fisher, Dispersion on a sphere, Proc. R. Soc. London, Ser. A 217, 295–305, 1953.